Lecture Notes: Regularization Part 2: Lasso (L1) Regression

This lecture is a follow-up to Ridge Regression. It assumes you are familiar with the concepts from that video [00:30].

I. Quick Review: Ridge (L2) Regression

1. **The Problem:** When we fit a model with **Least Squares** to a small training dataset, it can be **overfit**. This leads to a model with **low bias** (fits training data perfectly) but **high variance** (fails on new testing data) [01:38].
2. **The Ridge Solution:** Ridge Regression introduces a penalty to the cost function. It finds the line that minimizes:

$$SS_{Residuals} + \lambda \times (\text{Slope})^2$$

[01:54]
3. **The Result:** Ridge Regression accepts a **small amount of bias** (a slightly worse fit on the training data) in exchange for a **significant drop in variance**. This provides better long-term predictions [02:21].

II. Introducing Lasso (L1) Regression

Lasso Regression is extremely similar to Ridge Regression, with one critical change to the penalty [00:47].

- **Ridge (L2) Penalty:** Uses the **square** of the slope: $\lambda \times (\text{Slope})^2$
- **Lasso (L1) Penalty:** Uses the **absolute value** of the slope: $\lambda \times |\text{Slope}|$ [02:42]

The **Lasso Regression** cost function minimizes:

$$SS_{Residuals} + \lambda \times |\text{Slope}|$$

III. Similarities Between Ridge and Lasso

In practice, the two methods are very alike in many ways:

1. **Goal:** Both add a small amount of bias to the model to reduce variance, making predictions less sensitive to the training data [03:25].
2. **Lambda (λ):** The tuning parameter λ can be any value from 0 to positive infinity and is determined using **cross-validation** in both methods [02:53].
3. **Context:** Both can be applied to the same types of models (Linear Regression, Logistic Regression, models with discrete variables, etc.) [03:34].
4. **General Form:** For a complex model with many parameters ($\beta_1, \beta_2, \ldots$), the penalty includes all parameters *except* the y-intercept (β_0) [04:03].
   - **Ridge (L2):** $\lambda \times (\beta_1^2 + \beta_2^2 + \beta_3^2 + \ldots)$
   - **Lasso (L1):** $\lambda \times (|\beta_1| + |\beta_2| + |\beta_3| + \ldots)$

IV. The BIG Difference: Shrinking to Zero

The change from a *squared* penalty (L2) to an *absolute value* penalty (L1) has one massive consequence [05:00].

- **Ridge Regression (L2):** As you increase the value of λ, the parameters are forced to get smaller. However, because of the squared term, they will only get **asymptotically close to zero**. They will never *actually* reach zero [05:30].

- **Lasso Regression (L1):** As you increase the value of λ, the parameters are also forced to get smaller. But the absolute value penalty *allows* the parameters to be shrunk **all the way to zero** [05:25].

This means **Lasso can set a parameter's value to 0**, while Ridge can only make it 0.000001.

V. Practical Implication: Automatic Feature Selection

This ability to "zero-out" parameters makes Lasso a powerful tool for **feature selection** [07:01].

- **The Scenario:** Imagine a "big huge crazy equation" to predict `Size`, with many input variables (parameters) [05:47]:
  - **Good Variables:** `Weight`, `HighFatDiet`
  - **Useless Variables:** `AstrologicalSign`, `AirspeedOfSwallow` [05:58]
- **How Ridge Handles It:** When we apply Ridge Regression, increasing λ will shrink all the parameters. The parameters for `AstrologicalSign` and `AirspeedOfSwallow` will shrink *a lot*, but they will **never be exactly zero** [06:24]. Your final model still *technically* uses these useless variables.
- **How Lasso Handles It:** When we apply Lasso Regression, increasing λ will shrink the parameters for the useless variables **all the way to zero** [06:43].
  - The terms for `AstrologicalSign` and `AirspeedOfSwallow` literally "go away" [06:49].
  - The final model is simpler and *only* includes the useful variables: `Weight` and `Diet`.

VI. When to Use Which?

- **Lasso (L1):** Lasso is better at reducing variance when your model contains **a lot of useless variables**. It performs automatic feature selection and creates a simpler, easier-to-interpret model [07:01].
- **Ridge (L2):** Ridge Regression tends to do a little better when your model contains **mostly useful variables** that all contribute to the prediction [07:18].

VII. Summary

Feature	Ridge Regression (L2)	Lasso Regression (L1)
Penalty	Squares the parameters: $\lambda \sum \beta_i^2$	Takes the absolute value: $\lambda \sum
Effect	Shrinks parameters *asymptotically* to zero.	Shrinks parameters *all the way* to zero.
Best For...	Models where most variables are useful.	Models with many useless variables.
Main Benefit	Reduces variance.	Reduces variance **and** performs feature selection.
Final Model	Includes all variables.	Simpler, more interpretable, excludes useless variables.
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